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Viscoelastic phase separation in biological cells

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Biological phase separation forming membraneless organelles in cytoplasm and nucleus has attracted considerable attention. Liquid-like condensates are often created as spherical droplets. However, various condensates with network-like morphologies, including protein granules, localisation bodies, and centrosome assemblies, have recently been discovered in cells. Therefore, what controls the morphology of biological phase separation is a critical issue but remains elusive. Here, based on the knowledge of viscoelastic phase separation in soft matter physics, we propose that the difference in the molecular dynamics between the two phases controls the condensate morphology. Small and large mobility differences between the two phases should lead to droplet-like and network-like morphologies of the minority phase, respectively. We show that asymmetric partitioning of high-molecular-weight unstructured polymers (e.g., messenger RNA) between the two phases increases the dynamic asymmetry between the phases to form a network-like pattern of the slower phase, which may further be stabilised through inter-polymer binding.

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Recently, it has been widely recognised that membraneless organelles are formed by liquid-liquid phase separation (LLPS) in biological cells, and biological molecules compartmentalised in these domains control biological reactions. Soluble macromolecules, such as RNA (ribonucleic acid), proteins, and RNA protein complexes, can undergo phase separation to form membraneless condensates. Upon phase separation, some macromolecules are incorporated in the condensates, some are excluded from them, and some are neither included nor excluded and equally divided inside and outside the condensates. This spatial localisation of the components plays a crucial role in organising and activating biological functions. Many of these condensates have a spherical shape, including P granules, stress granules, and nucleolus[1–4]. This phase-separation morphology is typically observed when one phase occupies a much smaller volume fraction than another. The spherical domain shape is selected to minimise the interface free-energy cost. The multicomponent nature of biological systems[5] further leads to a variety of spherical droplets, such as core-shell and nested droplets, when more than three phases coexist[6].

Recently, it was also shown that some condensates age over time while slowing down the dynamics and becoming solid-like or gel-like states[7–12]. These aggregates lose their coalescence ability and tend to form network-like structures. Such network-like or irregular-droplet morphology has also been discovered in TIS (TPA (12-0-tetradecanoyl-phorbol-13-acetate)-inducible sequence protein) granules[13,14], RNP (ribonucleoprotein) granules[15], localisation bodies[16], FXR1 (fragile X mental retardation syndrome-related protein 1) condensates[17] and NLRP6 (nucleotide-binding oligomerization domain-like (NOD-like) receptor family pyrin domain containing 6)[18] (see also recent reviews[6,19]). This type of morphology provides a large interface area with a bicontinuous nature, crucial for biological reactions and transport.

Furthermore, solid-like properties of condensates are essential for supporting mechanical stress, e.g. upon the cell division[20]. For example, the centrosome organises microtubule arrays and comprises two centrioles surrounded by an amorphous protein mass called the pericentriolar material (PCM). It was recently shown[21] that large coiled-coil proteins polymerise and induce phase separation to form micrometre-sized interconnected, porous networks that specifically recruit downstream PCM proteins. Although the relevance of LLPS in centrosome assembly has been a matter of debate[22], a recent study by ref. [23] has supported the LLPS scenario. However, the origin of the unusual network-forming nature of phase separation has remained elusive.

In ordinary LLPS[24], the minority phase always forms droplet morphology to minimise the interface free-energy cost, which also drives domain coarsening. Droplet phase separation occurs through the nucleation-growth mechanism in a metastable state, whereas through spinodal decomposition in an unstable state. The sharpness of this border, i.e. the spinodal line, critically depends on the range of intermolecular interactions[24,25]. On the other hand, the network-like or mesh-like domain morphology of the minority phase is rather unusual and cannot be explained by the classical theory of LLPS. About 30 years ago, we found a special type of phase separation in polymer solutions[26,27], where even the minority phase can form a space-spanning network structure. We named this phase separation 'viscoelastic phase separation (VPS)' since viscoelastic effects play a dominant role in phase separation. Later, we have shown that VPS is also observed universally in various kinds of soft matter systems[28,29], including colloidal suspensions[30–32], protein solutions[33], and surfactant solutions[34]. The key to VPS is the dynamic asymmetry between the two components of a mixture. We emphasise that VPS is the only mechanism to form a space-spanning network structure of the minority phase, whose volume fraction is less than 30%. We note that the gelation of colloidal suspensions and protein solutions also belongs to VPS in the broad sense[30,31,33,35].

Based on VPS knowledge, this article explains how the domain morphology, droplet-like or network-like, is determined in biological phase separation. We show that the key is the mobility difference between the two phases to be formed. The network-like or mesh-like condensates can be formed via phase separation only when a significant mobility difference exists between the two phases. If the mobility difference is small, normal phase separation occurs. Since VPS is the most general form of phase separation, it can describe any type of phase separation, including the normal one[28,29,36]. Since biological cells are composed of a complex mixture of proteins, DNA (deoxyribonucleic acid), RNA, salt, water, and others, the phase separation behaviours can be pretty complex[6,37]. We note that water is always the component with the fastest dynamics in biological phase separation, playing a critical role in producing the dynamic asymmetry between the two phases. Nevertheless, we argue that the selective inclusion of large-size polymeric components such as RNA and DNA into one of the phases is responsible for causing a significant dynamic asymmetry between the two phases and inducing VPS. We demonstrate that this scenario can well explain recent experimental findings in mesh-like or network-like structure formation in TIS granules network[13,14], RNP (ribonucleoprotein) granules[15], localisation bodies[16], FXR1 condensates[17], NLRP6[18] and centrosome assemblies[21,23,38].

Results and discussion

Characteristics of viscoelastic phase separation. Let us consider phase separation into two phases with fast and slow dynamics. Below, we explain the essential physics of VPS as intuitively as possible without using any equations. We describe the formal theoretical frame of VPS[28,29] in the Methods section for readers interested in the theory, although it is unnecessary to understand the content of this article. We express the local polymer composition by $\phi(\mathbf{r})$ at point $\mathbf{r}$. Once phase separation is initiated, the composition gradient ($\nabla\phi(\mathbf{r})$) induced by the growth of the spatial composition fluctuations ($\phi(\mathbf{r})$) produces the growth of the velocity field ($\mathbf{v}(\mathbf{r})$), i.e. the deformation field (see Eqs. (3-5) in the Methods section). Note that the initial phase separation speed is determined by the average speed between the fast and slow components. Then, the faster phase can catch up with the deformation speed produced by phase separation and behaves as a fluid. However, the slower phase cannot follow the speed and behaves viscoelastically or elastically. In other words, the faster phase cannot support mechanical stress and is just low, whereas the slower phase can. As a result, the mechanical stress is generated selectively in the slow-component-rich phase and, thus, asymmetrically divided between the two phases. Therefore, the morphology of the slower phase is formed to satisfy the mechanical force-balance condition (see Eq. (5) in the Methods section). This mechanical pattern selection principle explains why network-like or mesh-like structures of the slower phase can be formed instead of droplet structures, even if it is the minority phase.

In the intermediate stage of VPS, the domain shape is determined by the mechanical force-balance condition instead of the interface area minimisation. From Eq. (5) in the Methods section, we obtain the following force-balance equation in a steady state:

$$\partial_i\left[K\left\{\partial_i\phi\partial_j\phi-\frac{1}{d}(\partial_i\phi)(\partial_j\phi)\delta_{ij}\right\}-\sigma_{Sij}+p\right]=0, \qquad (1)$$

where i and j are spatial coordinate labels, and δ_{ij} is Kronecker's

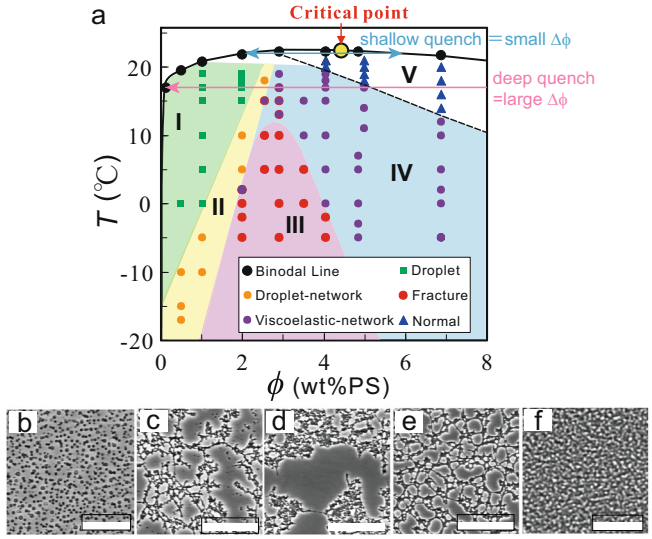


Fig. 1 State diagram and typical phase-separation patterns observed in a polystyrene/diethyl malonate mixture. a State diagram, in which we show what type of phase separation is observed at each state point. The weight-averaged molecular weight of polystyrene was $M_w = 7.04 \times 10^5$. ϕ is the concentration of polystyrene, and T is the temperature. Phase separation is induced by quenching the temperature from the one-phase region to a state point (see the symbols) in the two-phase region. The mixtures were sandwiched between two cover glasses, whose gap was set to 5 μm. A shallow/deep quench leads to a small/large difference in ϕ between the two phases, $\Delta\phi$ (see the blue/pink arrow), resulting in weak/strong dynamic asymmetry. The phase separation (PS) can be grouped into five types. (I): metastable-droplet PS; (II): droplet-network PS; (III): fracture PS; (IV): network PS; (V): normal PS. The dashed black line denotes the border between viscoelastic phase separation (VPS) and normal phase separation, i.e. the border between regions IV and V. **b** Typical phase separation pattern in region I. The scale bar corresponds to 50 μm. **c** Typical phase separation pattern in region II. The scale bar corresponds to 100 μm. **d** Typical phase separation pattern in region III. The scale bar corresponds to 400 μm. **e** Typical phase separation pattern in region IV. The scale bar corresponds to 200 μm. **f** Typical phase separation pattern in region V. The scale bar corresponds to 50 μm.

delta. Here K represents the coefficient associated with the interface tension (see Methods for the precise definition), $\boldsymbol{\sigma}$ is the shear stress, and p is the pressure. Equation (1) shows that the phase-separation morphology is determined to satisfy the balance between the interface force, the mechanical shear force, and the pressure force. However, in the late stage, the domain deformation rate generated by phase separation slows down since the system approaches the equilibrium two-phase coexisting state. Thus, even the slower phase can eventually follow the deformation rate and starts to behave as a fluid. In other words, the mechanical shear stress disappears, and the pattern tends to satisfy Eq. (1) with $\sigma_{Sij} = 0$. Then, the interface tension starts to determine the domain shape. For example, the interface force is balanced with the Laplace pressure force for droplet morphology. Thus, solidification, e.g. gelation and vitrification, is necessary to keep the network-like or mesh-like structures permanently.

As mentioned above, the network structure formed by VPS is under the mechanical force-balance condition (see Eq. (1)). In 2D and 3D, three and four force vectors generally realise the force balance, respectively. Thus, the natural number of arms at the junction point is typically 3 in 2D and 4 in 3D. Since polymers or proteins in the slower phase attract each other, the slower phase always tends to shrink its volume. Furthermore, the system tends to minimise the interface energy cost by decreasing the interface area. This volume-shrinking tendency and/or the interface energy minimisation means that the network arms are always under tension, i.e. stretched. Thus, the entire network tends to shrink[39]. Accordingly, the network's boundary should support the shrinking force to maintain the network structure. In other words, the boundary condition is a crucial requirement for VPS to form a network structure.

In biological cells, this can be achieved by the preferential wetting of the slower phase to the cell wall, nucleus, or membranes[40,41]. Recently, the importance of wetting effects on biological phase separation in cells has been shown[42,43]. For large biomolecular solutions, the interface tension can be very low[44], which may also impact the phase separation and wetting behaviours[45,46]. Here we point out that besides wetting-induced stabilisation, the network-like structure becomes stable even without the boundary once the structure is frozen by vitrification or gelation. It should be stressed here that vitrification or gelation does not necessarily mean that the slower phase is wholly frozen for phase separation of multi-component systems, which may be the case for biological systems. In multi-component systems, wetting allows some components in the network to keep molecular mobility without dynamic arrest even after vitrification or gelation of the network-forming phase (see below).

VPS in soft matter. Before discussing biological phase separation, we describe VPS characteristics observed in soft matter, such as polymer solutions, colloidal suspensions, and protein solutions. In ordinary soft matter experiments, phase separation is initiated by an instantaneous quench of the mixture temperature T from the one-phase to the two-phase region. Only when the temperature is quenched deeply enough to cause strong dynamic asymmetry between the two phases, can VPS be observed (see Fig. 1a).

First, we show typical patterns formed by VPS of polymer solutions in Fig. 1b–e. In region V (below the binodal line and above the black dashed line), ordinary phase separation described in the textbook[24] is observed because of the weak dynamic asymmetry due to the relatively small ϕ difference, $\Delta\phi$, between the two phases: When the two phases occupy similar volume fractions, the bicontinuous structure is formed (see Fig. 1f). When one phase occupies a much less volume fraction than the other, the minority phase forms droplet structures. However, this is not the case when the system is deeply quenched below the black dashed line in Fig. 1a, where the difference in ϕ between the two phases, $\Delta\phi$, is significant enough to cause a strong dynamic asymmetry between them.

In region I, the minority polymer-rich (more generally, the slower-component-rich) phase forms tiny droplets (see Fig. 1b). However, droplet coalescence is relatively rare due to the viscoelastic nature of droplets due to serious entanglements of dense polymers. The droplet collision and coalescence are characterised by two critical timescales[28]. These timescales are the characteristic time of the collision between two droplets (or the contact time), τ_c, and the characteristic rheological time of the polymer-rich phase, τ_t. Viscoelastic effects should play a role when $\tau_c < \tau_t$. The Brownian motion of a droplet with mass m is characterised by a randomly varying thermal velocity of magnitude $\langle v \rangle \sim (k_B T/m)^{1/2}$ and duration $\tau_r \sim mD_R/k_B T$ (D_R: the diffusion constant of a droplet with radius R). Thus, τ_c should satisfy the relation $r_{int}/\langle v \rangle < \tau_c < r_{int}^2/D_R$, where r_{int} is the range of interdroplet interaction. On the other hand, $\tau_t \sim \eta_s b^3 N^3 \phi^\alpha / k_B T$ (b: segment size; α: positive exponent) if we consider only topological entanglement effects. For typical values of the parameters, τ_t could be longer than τ_c for a large N or for a

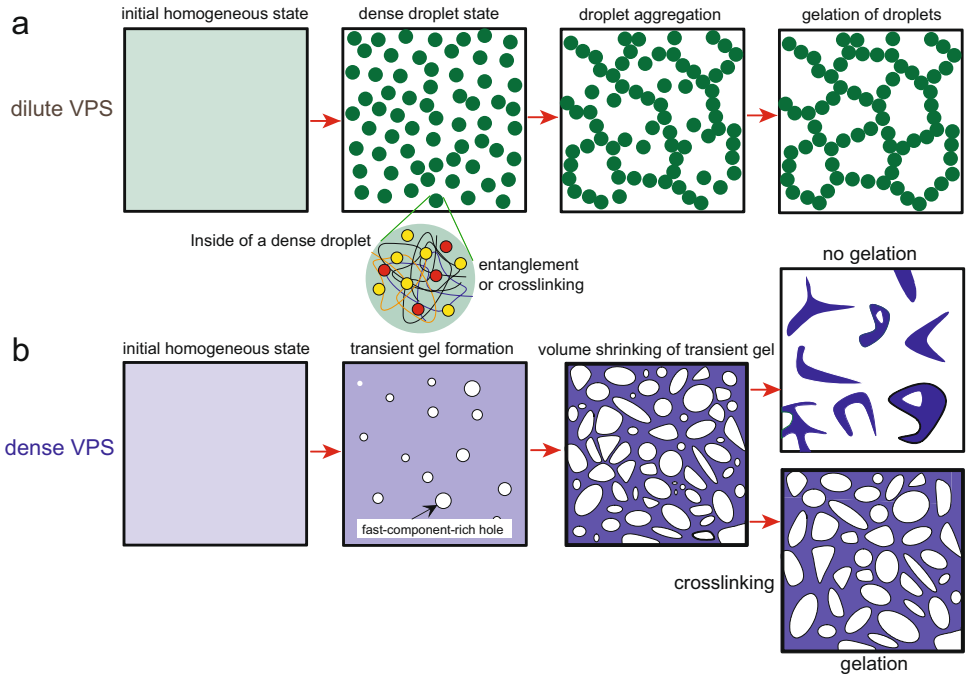


Fig. 2 Schematic representations of two typical types of viscoelastic phase separation (VPS). a VPS in region II of the phase diagram of Fig. 1a. Dense droplets composed of the slow component (polymer) are formed in the fast-component (solvent)-rich phase immediately after phase separation and behave like a gel ball due to the entanglement or crosslinking of polymeric components. We show an enlarged picture of a droplet as an example of biological phase separation, where yellow and red spheres represent binding proteins that crosslink polymers. Then, these droplets aggregate to form a percolated droplet network, i.e. a gel network. This type of VPS is observed when the slow-component-rich phase occupies a much smaller volume fraction than the fast-component-rich phase. **b** VPS in region IV of the phase diagram of Fig. 1a. A transient gel state is formed immediately after phase separation. Then, the transient gel shrinks while expelling the fast component (solvent) and forms a sponge-like structure. Its further volume shrinking transforms the structure from sponge-like to network-like. In the final phase separation stage, the domain deformation rate slows down drastically. Thus, the slower phase eventually behaves as a fluid, leading to the morphological transformation from network-like to droplet-like (no gelation). However, if the slower phase is dynamically arrested by crosslinking of polymeric components or vitrification, the sponge-like or network-like structure is frozen as a gel and stabilised (gelation). The darker domain colour in panels a and b represents higher polymer concentration.

deep quench, especially if energetic entanglements due to attractive interactions between chains, which makes τ_t much longer, are taken into account. Due to this mechanism of droplet coalescence, the droplet size R grows much slower than the ordinary droplet coarsening law of $R \sim t^{1/3}$ for simple liquid binary mixtures[24].

In region II, the minority polymer-rich phase forms droplets while expelling the solvent, and then droplets aggregate to form a percolated network (see Fig. 1c). During the initial droplet formation, the concentration of polymers inside droplets is rapidly increased while expelling the other less-viscous components (the solvent in a polymer solution). This process leads to the formation of droplets of strongly entangled dense polymers. This makes the interdiffusion of polymers between two contacting droplets extremely slow ($\tau_t \gg \tau_c$), and thus, interdroplet collision does not induce interface-tension-driven droplet fusion. This situation is markedly different from ordinary liquid-droplet collision ($\tau_c \gg \tau_t$), whose coalescence speed is controlled by γ/η (γ: the interface tension; η: the viscosity of the slower phase). In other words, dense polymer-rich droplets behave as gel balls, which further form a percolated network structure, similarly to colloidal gelation (here gel balls can effectively be regarded as colloids)[30–32,35]. The droplets can be stabilised further by inter-polymer crosslinking. VPS in region II is shown schematically in Fig. 2a. As examples of VPS in region II, we show the pattern-formation processes observed in a polymer solution and a protein solution in Fig. 3a–c and d–f, respectively. We can see the initial droplet formation followed by network formation. These network patterns are very similar to the mesh-like structure formed in TIS granules mentioned above[14] (see Fig. 3g–i) and porous network structures of centrosome assembly (see Fig. 4c, d in ref. [21]). As explained above, the force-balance condition (Eq. (1)) leads to the network morphology of the slow-component-rich phase, where three and four arms are connected at a junction in 2D and 3D, respectively, to satisfy the force-balance condition (see Fig. 3 for 2D).

In region IV, the polymer solution forms a transient gel state as a whole immediately after the initiation of phase separation. Then, solvent-rich holes are formed as droplets in the transient gel (see Fig. 2b). The transient gel keeps shrinking its volume with time while expelling the solvent into the solvent-rich holes, forming a space-spanning network structure (see Fig. 1e). However, in the very late stage, the network structure is broken up and transformed into droplet structures (see Fig. 2b; no gelation). This behaviour can be explained as follows. In the network-forming process, the polymer-rich phase with slow dynamics cannot catch up with the deformation speed induced by phase separation and behaves as an elastic body. Thus, the mechanical force-balance condition (see Eq. (1)) leads to the formation of a well-developed network structure of the slow-component-rich phase. In other words, the polymer-rich phase behaves as a gel. However, the system approaches the equilibrium state in the late stage, and thus, the domain deformation rate drastically slows down. Then, even the slow polymer-rich phase can catch up with the domain deformation and, thus, behaves like a liquid. Thus, the interface tension starts to play a role in pattern evolution, leading to the formation of spherical droplets. However, if gelation or vitrification arrests phase separation, the network pattern can be stabilised (see Fig. 2b; gelation).

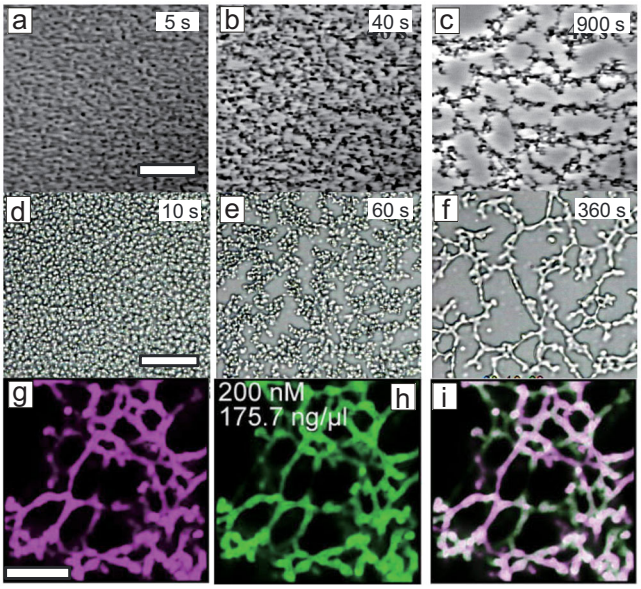


Fig. 3 Pattern evolution in viscoelastic phase separation (VPS). a–c A phase separation process observed in a mixture of polystyrene ($M_w = 7.04 \times 10^5$) and diethyl malonate (0.5 wt% polystyrene) at 20 °C with phase-contrast microscopy. The scale bar corresponds to 50 μm. **d–f** A phase separation process observed in a protein solution (protein concentration $c = 200$ mg ml^{-1}, salt concentration 7.5 wt%) at 37 °C with conventional optical microscopy. The scale bar corresponds to 50 μm. **g–i** Dynamic mesh-like condensates of TIS ((TPA (12-0-tetradecanoyl-phorbol-13-acetate)-inducible sequence protein)) granules formed in vitro[14]. Confocal images of phase separation experiments using purified mGFP (membrane-bound form of green fluorescent protein)-FUS (Fused in Sarcoma)-TIS (10 mM) in the presence of the indicated in vitro transcribed RNAs (ribonucleic acid) (CD274 3′UTRs (untranslated regions) (2075 nt); 200 nM, 175.7 ng μl^{-1}) after 16 h of incubation. **g** Image of FUS-TIS signal; **h** Image of RNA signal; **i** Merge of FUS-TIS and RNA signals. Five per cent dextran was added into the phase separation buffer as a crowding agent. The scale bar corresponds to 10 μm. These images in **g–i** are reproduced from the fourth column of Fig. 2a of ref. [14].

In region III, the effect of the mechanical stress stored in the shrinking polymer-rich phase becomes so intense that phase separation proceeds with mechanical fracture (see Fig. 1d)[47]. Here, we note that since the polymerisation reaction inducing phase separation may not be so fast in biological systems, the driving force of phase separation may not be strong enough to induce this fracture-type phase separation. This point needs to be confirmed.

In the above, we summarise what we know about the VPS of polymer solutions. In polymer VPS, there is no self-similar pattern growth because of the volume shrinking of the polymer-rich phase even after the sharp-interface formation[28,29]. However, the situation is very different for colloidal suspensions and protein solutions due to the absence of entanglements. It has been known empirically that for VPS in these systems, the characteristic domain size, i.e. the network pore size, R grows as $R \sim t^{1/2}$ [30–33,48]. We have recently confirmed the self-similarity of the phase-separation pattern by numerical simulations of colloidal phase separation, including hydrodynamic interactions, and revealed the physical mechanism behind this coarsening law[32]. Please refer to the paper by Tateno and Tanaka[32] for the detail of the mechanism.

Unique features of biological phase separation. Now, we discuss biological phase separation in the light of the knowledge of VPS described above. First, we consider the difference between phase separation in usual soft matter experiments and biological phase separation in cells. Phase separation in biological cells is usually not induced by a temperature change[24], but often by the polymerisation and/or the change of components, including proteins, polymers, salt, pH (potential of hydrogen), and ATP (adenosine triphosphate). Thus, the change of the thermodynamic condition inducing phase separation may be much slower than the instantaneous temperature quench. One of the essential mechanisms initiating phase separation in biological cells is polymerisation reaction[7,49–51]. Interactions between multivalent macromolecules such as proteins and RNA can induce polymerisation-driven phase separation. The Flory-Huggins free energy[52] (see Eq. (9) in the Methods section) tells us that the increase in the degree of polymerisation N leads to the decrease in

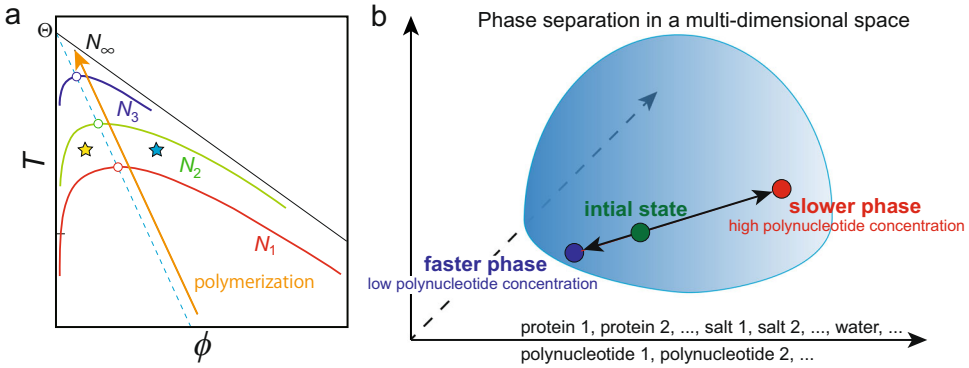


Fig. 4 Dynamic phase diagram of biological phase separation. a Polymerisation-induced phase separation. The phase diagram of a polymer solution moves towards the black line (N) with an increase in the degree of polymerisation N from N_1 through N_2 and N_3 ($N_1<N_2<N_3$). T is the temperature, and ϕ is the polymer concentration. With an increase in N, the critical point, (T_c, ϕ_c), where T_c and ϕ_c are the critical temperature and concentration, indicated by the open circle moves towards the critical point of the polymer solution of $N = \infty$, ($T_c = \Theta$, $\phi_c = 0$), where Θ is the Θ temperature[52]. The blue dashed line indicates the N-dependence of the critical point temperature T_c of the polymer solution. At the yellow star state point, the minority phase has much slower dynamics than the majority phase upon phase separation driven by polymerisation. In contrast, it is the opposite at the blue star position. The phase-separation behaviour also critically depends on the relationship between the polymerisation-reaction speed and the phase-separation speed. **b** Schematic phase diagram in a multi-dimensional space. In biological systems made of multi-components, phase separation always occurs in a multi-dimensional space. Multi-phase coexistence is possible, but we consider only a case of two-phase coexistence for simplicity. Upon phase separation, the components of a mixture, such as polynucleotides, proteins, and salts, are divided into two phases. The asymmetric partitioning of high-molecular-weight polymeric components into the two phases should produce a strong dynamic asymmetry between the two phases, leading to VPS.

the mixing entropy, increasing the phase-separation tendency. Thus, polymerisation generally induces phase separation when a system has an upper-critical-solution temperature-(UCST)-type phase diagram (see Fig. 4a).

However, we note that there is also a possibility that some factors other than polymerisation (e.g. salt concentration and pH changes) initiate phase separation, and condensation of RNAs and/or binding proteins triggers polymerisation. It is of critical importance to reveal whether polymerisation drives phase separation or is induced by phase separation.

Another characteristic feature of biological phase separation is the complex composition of the two phases. Polymer and protein solutions used in physics experiments are usually simple binary mixtures. On the other hand, in biological phase separation, a system is always made of multi-components, and both phases contain a variety of proteins and polymers[5]. Thus, phase separation takes place in a multi-dimensional space[37] even if the number of phases formed is two (see Fig. 4b). Both phases formed by phase separation contain various components in different concentrations, such as polynucleotides, proteins, salts, pH, and ATP. Thus, it is natural to expect that the difference in the concentrations of polymeric components such as RNA and DNA is critical for producing a large dynamic contrast between the two phases (i.e. dynamic asymmetry[28,29]) (see Fig. 4). In particular, polymerisation should drastically enhance the dynamic asymmetry between the two phases. Besides the dynamic factor leading to the network formation of condensates, it was also shown that the multi-component feature enhances the connectivity of condensates[53].

Mechanisms of dynamic arrest and solidification in biological phase separation. Here, we consider the origins of slow dynamics in biological phase separation. The glassy behaviour of condensates has been reported recently[8–12]. In particular, ref. [12] have succeeded in observing the viscoelastic relaxation in condensates directly and estimating the relaxation time τ_S. Glassy slow dynamics can generally be induced by dense packing of particles or molecules. In this context, it is critical to note that VPS is generally accompanied by condensation of slow components during phase separation[28,29]. In ordinary phase separation, after the sharp interface is formed between the two phases, the concentrations in the two phases already almost reach the final equilibrium concentrations determined by the lever rule of the thermodynamic phase diagram[24]. In contrast, this is not the case for VPS containing polymer components in regions IV and III of Fig. 1a[28,29]. Even after forming a sharp interface, the polymer-rich slower phase keeps shrinking its volume while increasing the local concentration, further slowing down the dynamics of the slower polymer-rich phase. This condensation process is critical when the slower phase contains polymeric components, such as RNA and DNA.

Next, we consider mechanisms leading to the dynamic arrest of VPS, i.e. solidification of the network structure. If there are strong enough attractions between polymers, they form a gel and stabilise a network-like structure. On the other hand, for colloidal suspensions and protein solutions, the dynamic arrest can be induced by vitrification of the dense phase[54,55] or percolation of mechanically rigid isostatic local structures[56,57]. These dynamic arrest mechanisms can explain viscoelastic effects on LLPS, including gelation, in homotypic phase separation[58], i.e. for single-component condensates (like FUS). For biological phase separation, a polymerisation mechanism by binding proteins is also an essential mechanism of dynamic arrest, as described below. We also note that even when both phases include big molecules similarly, strong dynamical asymmetry can be caused by polymerisation (or gelation) only in one of the phases or by the significant difference in the glass-transition temperature between the two phases[27].

In relation to the above, we mention a few interesting observations on the dynamic arrest of biological phase separation. For example, Wang et al.[59] reported that proteins, such as FUS, phase separate to form liquid-like condensates that can harden toward less dynamic structures. Furthermore, they showed that glycine residues enhance fluidity, whereas glutamine and serine residues promote hardening. This study clarified how a sequence-encoded molecular grammar underlies the driving forces of phase separation of proteins in the FUS family. Such information may provide a clue to the molecular-level understanding of the morphology control of phase separation. Many nuclear and cytoplasmic condensates are also rich in RNA and RNA-binding proteins. Recently it was shown that they play a critical role in LLPS[60]. Lin et al.[7] showed that droplets mature to more stable states over time, which was assessed by slowed fluorescence recovery after photobleaching and resistance to salt. Such maturation is often brought by the formation of fibrous structures, leading to the transformation of dynamic liquid droplets to more solid-like ones.

Polymerisation-induced phase separation and viscoelastic effects. Here we briefly describe the importance of polymerisation-induced viscoelastic effects on biological phase separation. To describe the polymerisation-induced phase separation, we have to describe the temporal change in the degree of polymerisation N. The simplest way to describe $N(t)$ is to assume the following time dependence of N (t):

$$N(t) = N(\infty) - (N(\infty) - N(0))\exp(-\Gamma t), \qquad (2)$$

where $N(0)$ and $N(\infty)$ are $N(t)$ at $t = 0$ and $t = \infty$, respectively, and Γ is the polymerisation rate. This temporal change in N leads to the changes of the phase diagram through Eq. (9) in the Methods section as well as viscoelastic properties. Combining this $N(t)$ with the set of the equations describing VPS (see Methods), we can describe polymerisation-induced VPS (see, e.g. ref. [61]).

Then, the phase-separation behaviour critically depends upon the polymerisation-reaction rate Γ and the rate of phase separation. Polymerisation reaction increases the dynamic asymmetry between the two phases necessary to induce VPS. For VPS to be induced, strong viscoelastic effects due to the dynamic asymmetry between the two phases must become active much before the concentrations reach the equilibrium values. If the polymerisation is faster than phase separation, viscoelastic effects should come into play and lead to VPS. Even the minority phase can form a network structure in this case. Here we note that when polymers such as RNAs and DNAs have enough high-molecular weights, VPS can occur even without polymerisation.

On the other hand, viscoelastic effects play only a minor role in the opposite case, i.e., when the polymerisation rate is slower than the phase-separation rate, since the dynamic asymmetry may not be enough to cause VPS: The minority phase tends to form droplets, as in ordinary phase separation. Even in such a case, unusual features can be observed. For example, the number of droplets and their volume fraction may increase with time, reflecting the temporal change in the phase diagram (see Fig. 4a). Such behaviours have indeed been observed in phase separation of the autophagy-initiating Atg1 (AuTophaGy related 1) complex[62]. We note that, in ordinary phase separation, the number of droplets $n_{droplet}$ monotonically decreases (typically, $n_{droplet} \sim t^{-1}$), and their volume fraction is constant with time. We also mention that the polymerisation and the resulting difference in the viscoelastic behaviour between the two phases

suppress droplet coalescence and thus slow down the domain coarsening rate. There is also a possibility of double phase separation[63].

We also note that a recent study has shown that gelation can be induced by the configurational entropy gain of interchain bindings over intrachain ones[64]. Thus, we need to consider both energetic and entropic factors in binding-induced gelation.

Reversibility of VPS and solidification. Here, we mention the reversibility of phase separation and solidification in biological phase separation. Since VPS is induced by dynamic effects, it must be reversible: Once the driving force of phase separation is removed by the change in environmental conditions, such as temperature, salt, pH and ATP, condensates should resolve and return to a homogeneous fluid state. If the dynamic arrest is brought by vitrification[54,55] or percolation of mechanically rigid local structures[56,57], solidification should be reversible since vitrification and cluster formation are reversible. Even a gel state formed by polymerisation in biological systems should be reversible since polymerisation is not due to chemical reactions modifying chemical bondings but through reversible physical reactions such as electrostatic or hydrogen bondings. This reversibility of VPS, including gelation, may be critical for organising biological functions, such as cell divisions (see below).

Formation of mesh-like condensates in TIS granules. Here we consider the formation of mesh-like condensates in TIS granules reported by Ma, Mayr, and their coworkers[13,14]. First, we summarise their experimental findings. TIS granules form through the assembly of the RNA-binding protein TIS11B (TPA-inducible sequence 11B). It was speculated that the network structure is formed by gelation[13]. However, it has recently been found by FRAP (fluorescence recovery after photobleaching) experiments[14] that TIS granules have dynamic protein components, similarly to L-bodies in frog oocytes[16]. This finding apparently indicates that the network still retains the liquid-like nature. It has been thought that biomolecular condensates with dynamic protein components should generally form spherical droplets to minimise the interfacial free-energy cost. So, the fundamental question is why the mesh-like structure can be formed.

Ma et al. concluded that mRNAs (messenger RNAs) play a critical role in determining the mesh-like morphology of condensates, based on their finding that the RNA-binding domain (RBD) in TIS11B is essential for this morphology formation. Furthermore, the large disordered regions of RNAs are necessary to form mesh-like structures; in the absence of these features, spherical droplets are formed. For example, RNAs tending to form intrachain local structural ordering cannot form mesh-like structures and forms spherical droplets. It was shown that intermolecular RNA–RNA interactions mediated by the large disordered regions of mRNAs are essential for promoting network formation.

They also found that RNA dimerisation elements enhance the formation of mesh-like structures in living cells. Based on these observations, they concluded that phase separation of an RNA-binding protein, together with primarily structured RNAs, generates sphere-like condensates, whereas phase separation of the same RNA-binding protein bound to RNAs capable of generating an extensive network forms mesh-like condensates in living cells. Furthermore, they reported slower motion of network-forming RNAs than sphere-forming RNAs in the condensates using the FRAP experiment (Fig. 7F of ref. [14]), indicating a stronger dynamic asymmetry between the two phases for the former than the latter.

They also studied the early stage of phase separation. Condensates that form droplets fuse upon collision, consistent with the Brownian-coagulation mechanism[24]. On the other hand, in the case of mesh-like structure formation, when two condensates that contain RNAs with large disordered domains come into contact, they do not fuse but connect at the contact sites. This enables the linkage of individual condensates, leading to mesh-like network structures (see Fig. 3g–i). This network-forming pathway is strikingly similar to the phase separation of a polymer solution in Region II of Fig. 1a (see also Fig. 3a–c) and a protein solution (see Fig. 3d–f).

Physical mechanism of mesh-like structure formation in biological systems. Recent studies on phase separation forming mesh-like structures in TIS granules network[13,14] (see above), RNP (ribonucleoprotein) granules[15], localisation bodies[16], FXR1 condensates[17], and NLRP6[18] all indicate the importance of unstructured RNAs promoting intermolecular RNA–RNA interactions in producing networks. Longer RNAs have more sites for intermolecular RNA–RNA interactions[65]. This mechanism provides another reason for long RNAs to enhance dynamic asymmetry and cause network-like phase separation[13,14], besides causing strong entanglements. In the light of VPS, we argue that such RNAs play a vital role in inducing the strong dynamic asymmetry between the two phases: The phase including RNAs and multivalent RNA-binding proteins has much slower dynamics than the other phase (see above and Fig. 7F of ref. [14]), forming mesh-like structures. It was shown[14] that RNAs with large disordered regions have a high propensity to induce network formation (see Fig. 3g–i). This behaviour can be explained as follows. Structured RNAs tending to form compact structures cannot induce slow enough dynamics. In contrast, unstructured RNAs can lead to slow dynamics due to severe entanglements and produce significant dynamic asymmetry between the two phases. The crosslinking induced by multivalent RNA-binding proteins also helps further slow down and enhancement of the elastic nature of tiny droplets that are to be connected to form network structures, similarly to VPS in region II (see Figs. 1a, 3).

Here we consider whether VPS observed in region IV is relevant to biological phase separation. Only the difference between regions II and IV is the volume fraction of the slower phase (see Figs. 1a, 2). In region II, the network structure is formed in two steps, i.e., the formation of tiny droplets with slow internal dynamics followed by their aggregation to form networks (see Fig. 2a). This latter process is essentially the same as the network formation in VPS of colloidal suspensions[31,32,35,57] by regarding polymer-rich tiny droplets as colloids. Hydrodynamic interaction between tiny droplets plays a critical role in forming a space-spanning network, i.e. the percolation of droplets[31,32,48]. The network structure formed by this type of VPS has a weaker volume-shrinking tendency since the volume shrinking already takes place when the tiny droplets are formed. Nevertheless, the boundary condition helps stabilise the network structure. We note that in the TIS granules, the network structure wets the nucleus (see Fig. 1 of ref. [14]) and probably the cell wall.

In relation to the mechanical stress generated in the network structure, we note that there are two types of network formation. In VPS-type, a network is formed by the dynamical arrest of phase separation. In this case, the percolation of the phase-separated structure occurs before the dynamical arrest, generating mechanical stress in the network due to the volume-shrinking tendency. We have recently found another type in dilute colloidal suspensions[66], in which percolation occurs after the local dynamical arrest, i.e. the formation of mechanically stable, rigid clusters. In this case, topological percolation generates little

mechanical stress, and the resulting network is almost stress-free when it is formed. We also show that the selection of these two types of gelation (stressed and stress-free) in colloidal suspensions is determined solely by the colloid volume fraction as long as the interaction is short-ranged[66]. Judging from the network structures reported in ref. [14] (see Fig. 3g–i), the network formation in TIS granules belongs to the VPS-type phase separation in region II.

In region IV, on the other hand, a transient gel state is formed in the beginning and starts to form a network structure accompanying volume shrinking (see Fig. 2b). Thus, strong mechanical stress is generated in the network due to its volume shrinking. In this type of VPS, the boundary condition plays a critical role in maintaining the network structure[39]. Although the network structure eventually transforms to the droplet one for this type of VPS, gelation by crosslinking of polymers or vitrification can stabilise the network structure (see Fig. 2b, lower path). It is interesting whether this type of VPS occurs in biological cells. This type of phase separation forms a sponge-like morphology in 3D. It is interesting to study whether the low-density sponge-like structures observed in Cajal Bodies, nucleoli, and speckles in the Xenopus Oocyte nucleus[67] are formed by this type of VPS or not. We note that the sponge-like or network-like structures formed by VPS provide not only a large surface area directly contacting the outer phase, essential for biological reactions, but also a percolated fluid channel allowing efficient material transport in the solid network.

The importance of polymeric components, such as RNA and DNA, in producing slow dynamics or dynamic arrest can also be seen in many cases. For example, it was reported for mitochondrial nucleoids that the addition of long polymerised strands of DNA, irrespective of the sequence, leads to favourable interactions between mtTFA (mitochondrial transcription factor A) and DNA, and increasing DNA leads to more irregular-droplet morphologies[68]. Also, for LLPS of histone proteins in cells, the irregularity of droplet shape is enhanced with the presence of polynucleocido[69]. It was also shown that viscoelastic properties can be controlled by polypeptides and RNA sequences as well as their mixture compositions[70].

Whether the phase has slow dynamics due to entanglement effects (i.e. very viscous but a fluid) or is dynamically arrested by crosslinking (i.e. forms gel) determines the final phase-separation morphology. For the former, the shape should be eventually determined by the interface free energy and thus becomes spherical at a long-time limit. On the other hand, for the latter, the irregular-droplet shape or sponge-like structures can be regarded as practically permanent in the timescale of biological cell activity.

Importance of the elasticity of condensates in biological functions. Here, we note that the solid-like nature of condensates should also play a critical role in biological processes where mechanics is critical, e.g. cell divisions[20]. Phase-separated condensates may allow the formation of an ordered microtubule array at the centrosome, and their solidity is crucial for supporting stresses and forces at the centromere and within the microtubule spindle upon cell divisions. It was recently shown[21,23] that large coiled-coil proteins polymerise and induce phase separation to form micrometre-sized interconnected network structures, which is the critical process of centrosome assembly. For example, Woodruff suggested the following scenario:[38] The mitotic pericentriolar material (PCM) scaffold initially assembles through LLPS of centrosome scaffold proteins around the centrioles due to multivalent interactions between coiled-coil domains to form spherical droplets, and then droplets rapidly solidify, transforming the PCM into a porous, gel-like material through linking solid-like droplets. The elasticity of PCM allows supporting microtubule-mediated pulling forces. This process is precisely the process of VPS in region II of Fig. 1a, which can be confirmed, e.g. by comparing Fig. 4c, d of ref. [21] with the network patterns in Fig. 3. Thus, VPS provides a realistic scenario for forming elastic network structures that can sustain mechanical stress upon cell divisions.

Impact of the intermolecular interaction range on viscoelasticity and network-like structure formation. Next, we briefly consider the formation of solid aggregates in biological phase separation and its relation to the range of the intermolecular interactions. Biomolecular condensates are usually in a liquid state but can be solid aggregates in neurodegenerative diseases or under environmental stresses (e.g. UV (ultraviolet) exposure, energy depletion, and pH change)[2,71–74]. Thus, a critical question is what molecular features cause solid aggregates without polymerisation reactions[2,71,75]. Protein molecules are often composed of two internal structures: poorly-folded "polymer-like" domains (intrinsically disordered regions, IDRs) and highly-folded "colloid-like" domains. So far, the polymer-like feature of proteins has mainly been considered to understand biomolecular condensates[2,4,71,73,75]. However, the importance of the colloid-like feature of proteins has recently been recognised[76]. For example, ref. [74] showed that native prion proteins (with IDRs) undergo liquid phase separation in living cells, whereas the removal of IDRs from prion proteins leads to gelation. We may regard polymer-like IDRs with charged domains as the sources of long-range interactions and the gelling mutants as colloids with short-range interactions. Weber[77] also showed that the sequence of nuclear bodies' components encodes mechanical properties: Proteins with high ratios of charged/aromatic amino acids assemble into liquid-like droplets that concentrate enzymes to enhance their biochemical activity, such as mRNA processing. On the other hand, proteins with low ratios of charged/aromatic amino acids (green) assemble into solid-like gels or amyloids that sequester enzymes to prevent them from acting on their targets. This relationship between charge density and fluidity also suggests the link of the interaction range to the mechanical properties of condensates.

We have recently found that colloids with short-range attractions should directly form gels upon phase separation without passing through a liquid state[66], whereas colloids with long-range interaction (e.g. Lennard–Jones interactions) phase separate without gelation, and the network structure continues to coarsen[32]. We stress that the short-range interaction is essential to forming packing-induced rigid mechanical structures. On noting the similarity between colloid and globular protein phase separation[33], this finding could help understand the physical mechanism of biomolecular condensates of solidity in terms of the range of intermolecular interactions.

Role of electrostatic interactions on phase separation. In the above, we mention that electrostatic interactions enhance the fluidity of condensed phases due to their long-range nature. Since biological polymers are electrolytes, electrostatic interactions are one of the critical interactions controlling phase separation (see, e.g. ref. [72]). Electrostatic interactions are repulsive for like charges, which can influence the phase separation largely. Attractive interactions between like species drive phase separation. Thus, repulsive interactions between like species may frustrate phase separation. For example, competition between short-range attractions and long-range repulsions is known to produce various interesting mesoscopic phase-separation patterns[78]. For

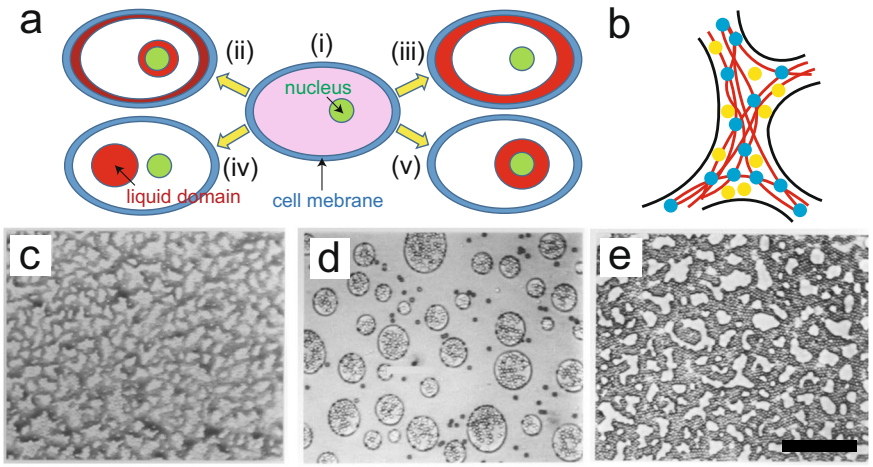


Fig. 5 Wetting effects on biological phase separation. a Schematic picture showing the possible wetting effects on phase separation in a cell surrounded by the cell membrane and containing a nucleus. Here we consider only the cell wall (blue) and nucleus (green) as elastic components for simplicity. The pink liquid confined in the cell (image (i)) phase separates into the red and white liquid phases. For such a case, there can be four possible situations. Image (ii): The red liquid phase wets both the cell membrane and nucleus. Image (iii): The red liquid phase wets only the cell membrane. Image (iv): The white liquid phase wets both the cell membrane and nucleus. Image (v): The red liquid phase wets only the nucleus. **b** Schematic picture of coexistence of immobile polymers (red) connected by binding proteins (blue) and mobile proteins (yellow) in a network structure. **c** Phase separation in a binary liquid mixture (oligo-caprolactone/oligo-styrene (3/7)) containing immobile glass beads (diameter = 7 μm and volume fraction = 9%). The phase-separation pattern was observed after 600 s from a temperature quench from 140 to 100 °C. **d** Phase separation in the same binary liquid mixture containing mobile glass beads (diameter = 7 μm and volume fraction = 3.2%). The phase-separation pattern was observed after 1710 s from a temperature quench from 140 to 60 °C. **e** Phase separation in the same binary liquid mixture containing mobile glass beads (diameter = 7 μm and volume fraction = 19%). The phase-separation pattern was observed after 3180 s from a temperature quench from 140 to 60 °C. The high particle density confined in the more wettable phase makes the dynamics glassy. The black bar corresponds to 100 μm.

example, we found in colloidal suspensions that frustration leads to multiple arrested states[79]. Similarly, droplet coalescence may be prevented by repulsive interactions between droplets while keeping the fluidity of droplets instead of (visco-)elastic stabilisation of droplets. The role of electrostatic interactions on biological phase separation is an interesting topic for future research.

Wetting effects on biological phase separation. Finally, we consider wetting effects[40] on biological phase separation, which is critical for biological systems. First, we consider the confinement, or boundary, effects on biological phase separation. Biological phase separation occurs inside a tiny biological cell, surrounded by cell membranes and containing various elastic components, such as a nucleus, mitochondrion, microtubules, endoplasmic reticulum, and actin filaments. Thus, phase separation is inevitably affected by preferential wetting to these elastic components. For example, wetting effects on elastic components, such as cell walls, are crucial for supporting the mechanical stress generated by VPS, as discussed above. For normal liquid-type phase separation, wetting effects affect the domain shape and location of domains formed by phase separation[42,43,80]. In Fig. 5a, the pink phase separates into the minority (red) and majority (white) phases. Depending on the wettability of the two phases to cell membranes (blue wall) and a nucleus (green sphere) components of a cell, liquid domains of the minority (red) phase can have four types of specific spatial localisation, as shown in Fig. 5a.

However, there are other elastic components in a cell, such as mitochondria, microtubules, endoplasmic reticulum and actin filaments. The domain coarsening may be suppressed by the wetting to these elastic components. Even in such a case, slow coarsening due to the evaporation-condensation mechanism may proceed if the interface curvature is inhomogeneous. If elastic components are spatially fixed in the cell, they trap liquid domains, as shown in Fig. 5c. In such a case, the domain shape may be affected by the spatial distribution of pinning objects and become non-spherical[41,81].

There is another type of wetting effect when elastic components are mobile[41,81]. Some components are directly responsible for inducing phase separation through the interaction between the like components (see the χ-parameter in Eq. (9) in the Methods section). However, some passive elastic components may not be a direct cause of driving phase separation but are still divided between the two phases by the difference in the wettability between them (see Fig. 5b)[41,81], as described above. They are preferentially included in the more wettable phase, as shown in Fig. 5d, e. Such preferential inclusion of elastic components into one phase may slow down the domain coarsening. When the packing density of elastic components becomes very high (see Fig. 5e), the phase containing them becomes glassy even if the phase containing them is a non-viscous fluid. Although relatively large-size glass beads were used in these experiments, the basic features should not depend on the particle size and shape. Thus, the above scenario for glass beads should straightforwardly apply to the case of much smaller objects, e.g. proteins. When particles are equally wettable to the two phases, they are localised at the domain interface, preventing further coarsening, similarly to the Pickering emulsion[82]. Such behaviour is indeed observed in biological phase separtion[83].

Based on the above knowledge, we consider why mesh-like condensates formed in cells contain highly mobile protein components. Ma et al.[14] showed that FUS-TIS and SUMO-SIM-TIS are mobile in the filamentous networks, suggesting that these protein components are not aggregated. We speculate that these mobile proteins have high wettability to the slower phase but are not tightly involved in gel-like structures, making the coexistence of gel-like elastic properties of the network structure and liquid-like properties of some proteins possible (see Fig. 5b). Another possibility is that proteins are attached to polymeric components' flexible parts (e.g. free ends). The inclusion of

mobile particles (e.g. proteins) into the solid-like phase via wetting while maintaining mobility may play a crucial role in biological functions.

Conclusions

Here we have provided fresh physical insight into biological phase separation from the perspective of viscoelastic phase separation in soft matter physics. Although phase separation in biological cells is far more complex than in soft matter, the fundamental physical nature should be the same. We have shown that the phase-separation morphology of the minority phase can be either droplet-like or network-like patterns, and its selection is determined by the dynamic asymmetry between the two phases. Only when the dynamic asymmetry is strong enough between the two phases, can the network structure of the minority phase be formed. Experimental results showing network-forming phase separation in biological cells have indicated that unstructured polymeric components such as mRNA and DNA play a critical role in the network-forming phase separation. We have shown that this feature is essential for creating the strong dynamic asymmetry between the two phases and thus VPS. It is interesting to follow the pattern-formation kinetics during biological phase separation and the temporal change of the two phases' molecular dynamics and rheological properties. Such a study is highly desirable for distinguishing VPS in regions II and IV (see Fig. 2). Polymerisation reactions should also affect the domain coarsening even for droplet-type phase separation. Characterising the polymerisation reaction is crucial for elucidating the role of polymerisation in phase separation. In this work, we focus on the network-like pattern formation of the minority phase. Recently, Dufresne and his coworkers[84–86] studied the elastic effects on phase separation in a system where the viscoelastic or elastic phase is the majority. The relevance of this type of phase separation in biological cells is also an interesting topic for future research. It will also be important to reveal the relationship between phase-separation morphologies (droplet-like or network-like) and biological functions and the mechanism by which biological systems select the morphologies.

Our work is a preliminary step toward understanding biological phase separation, which can be influenced by viscoelastic effects due to the asymmetric division of polymers and proteins into the two phases. We hope that our work facilitates further detailed studies of the impact of viscoelasticity and dynamic asymmetry on biological phase separation.

Methods

Theory of viscoelastic phase separation. Here, we briefly explain the viscoelastic model of phase separation[24,28,29] and the physical meaning of the fundamental equations as intuitive as possible. If a reader is not interested in the theory, this section can be skipped.

The fundamental equations describing VPS of a polymer solution are given as follows[28,29]:

$$\frac{\partial \phi}{\partial t} = -\nabla \cdot (\phi \mathbf{v}) + \nabla \cdot \frac{\phi(1-\phi)^2}{\zeta} \nabla \cdot [\overleftrightarrow{\mathbf{\Pi}} - \overleftrightarrow{\boldsymbol{\sigma}}]. \tag{3}$$

$$\mathbf{v}_\mathrm{p} - \mathbf{v} = -\frac{(1-\phi)^2}{\zeta} \nabla \cdot [\overleftrightarrow{\mathbf{\Pi}} - \overleftrightarrow{\boldsymbol{\sigma}}]. \tag{4}$$

$$\rho_0 \frac{\partial \mathbf{v}}{\partial t} = -\nabla \cdot [\overleftrightarrow{\mathbf{\Pi}} - \overleftrightarrow{\boldsymbol{\sigma}}] - \nabla p + \eta_s \nabla^2 \mathbf{v}. \tag{5}$$

$$\nabla \cdot \mathbf{v} = 0. \tag{6}$$

Equation (3) describes the convective diffusion equation, which shows that the local polymer composition $\phi(\mathbf{r}, t)$ can change through both diffusion and hydrodynamic transport by the flow. $\mathbf{\Pi}$ is the osmotic stress tensor, which is related to the thermodynamic force $\mathbf{F}_\phi$ as

$$\mathbf{F}_\phi = -\nabla \cdot \overleftrightarrow{\mathbf{\Pi}} = -\phi \nabla (\delta \mathscr{F}\{\phi(\mathbf{r})\}/\delta \phi(\mathbf{r})), \tag{7}$$

where $\delta/\delta\phi(r)$ represents the functional derivative in terms of $\phi(\mathbf{r})$. Here, $\mathscr{F}\{\phi(\mathbf{r})\}$ is the free energy functional of the polymer solution and given by the following equation[52] according to the Flory-Huggins theory, which was shown to be relevant to describing the thermodynamics of biological phase separation[75]:

$$\mathscr{F}\{\phi(\mathbf{r})\} = \int d\mathbf{r}[f_\mathrm{p}(\phi(\mathbf{r})) + \frac{1}{2} K |\nabla \phi(\mathbf{r})|^2], \tag{8}$$

$$f_\mathrm{p}(\phi(\mathbf{r})) = \frac{1}{N} \phi(\mathbf{r}) \ln \phi(\mathbf{r}) + (1-\phi(\mathbf{r})) \ln(1-\phi(\mathbf{r})) + \chi \phi(\mathbf{r})(1-\phi(\mathbf{r})), \tag{9}$$

where K is a positive constant suppressing the inhomogeneity of $\phi(\mathbf{r})$, N is the degree of polymerisation of polymers, and χ is the interaction parameter leading to the phase separation. The thermodynamic force $\mathbf{F}_\phi$ arises from the composition gradient $\nabla\phi$, i.e. the interface tension, for the incompressible fluid. In addition to the thermodynamic force, there is the mechanical force $\mathbf{F}_\sigma = -\nabla \cdot \overleftrightarrow{\boldsymbol{\sigma}}$, where $\overleftrightarrow{\boldsymbol{\sigma}}$ is the mechanical stress tensor. This mechanical force is absent in ordinary LLPS but plays a critical role in VPS. Unlike ordinary phase separation, the relative motion of polymers is driven by not only the thermodynamic osmotic stress $\overleftrightarrow{\mathbf{\Pi}}$ but also the mechanical stress $\overleftrightarrow{\boldsymbol{\sigma}}$ under the polymer-solvent friction (ζ: the friction constant per unit volume) (see Eq. (4)). $\mathbf{v}$ is the average velocity of a mixture given by $\mathbf{v} = \phi \mathbf{v}_\mathrm{p} + (1-\phi)\mathbf{v}_\mathrm{s}$, where $\mathbf{v}_\mathrm{p}(\mathbf{r}, t)$ and $\mathbf{v}_\mathrm{s}(\mathbf{r}, t)$ are respectively the coarse-grained average velocities of polymer and solvent at point $\mathbf{r}$ and time t. It transports $\phi(\mathbf{r}, t)$ via flow (see the first term of the right-hand side of Eq. (3)).

Equation (5) describes the momentum conservation, or the force balance, among the thermodynamic, mechanical, pressure, and viscous forces. ρ_0 is the average density, η_s is the solvent viscosity, and p is a part of the pressure, which is determined to satisfy the incompressible condition (Eq. (6)).

The terms containing the mechanical stress tensor cause the couplings between the composition (ϕ) and the stress fields ($\overleftrightarrow{\boldsymbol{\sigma}}$) via the velocity field ($\mathbf{v}$). The role of the mechanical force can be easily understood by considering a gel case. To close these equations, we need a constitutive equation, which describes the time evolution of $\overleftrightarrow{\boldsymbol{\sigma}}$.

For simplicity, in the above, we considered a case of polymer solution, where only polymers can support viscoelastic stress. For a more general case, where viscoelastic stress is not supported by just one of the components, we need a more general set of equations. In such a case, the constitutive relation may also become more complex[36]. Nevertheless, the above theory may be enough to capture the essence of VPS, including biological phase separation.

Next, we consider how the mechanical stress, $\overleftrightarrow{\boldsymbol{\sigma}}$, should be expressed in the case of a polymer solution. In general, we should incorporate relevant constitutive equations into the above two-fluid model, depending upon the type of material. Here, we employ the simplest upper-convective Maxwell-type constitutive equation:

$$\frac{D}{Dt} \overleftrightarrow{\boldsymbol{\sigma}}_\mathrm{S} = \overleftrightarrow{\boldsymbol{\sigma}}_\mathrm{S} \cdot \nabla \mathbf{v}_\mathrm{p} + (\nabla \mathbf{v}_\mathrm{p})^T \cdot \overleftrightarrow{\boldsymbol{\sigma}}_\mathrm{S} - \frac{1}{\tau_\mathrm{S}(\phi)} \overleftrightarrow{\boldsymbol{\sigma}}_\mathrm{S} + G_\mathrm{S}(\phi)\{\nabla \mathbf{v}_\mathrm{p} + (\nabla \mathbf{v}_\mathrm{p})^T\}, \tag{10}$$

where the Lagrange derivative $\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v}_\mathrm{p} \cdot \nabla$, and τ_S and G_S are the relaxation time and the modulus of the shear stress, respectively. Note that $(\nabla v_\mathrm{p})_{ij} = \partial_i v_{pj}$. To make the shear stress a traceless tensor, $\overleftrightarrow{\boldsymbol{\sigma}}_s$ is defined as $\overleftrightarrow{\boldsymbol{\sigma}}_s = \overleftrightarrow{\boldsymbol{\sigma}}_s - \frac{1}{d} \mathrm{Tr} \overleftrightarrow{\boldsymbol{\sigma}}_s \overleftrightarrow{\mathbf{I}}$, where $\overleftrightarrow{\mathbf{I}}$ is the unit tensor and d is the space dimensionality ($d = 3$ for ordinary biological phase separation and $d \sim 2$ for phase separation in a membrane).

We also need to consider the bulk stress in addition to the shear stress to describe the volume-shrinking behaviour of VPS[28,29]. This bulk stress expresses the spatial connectivity of a transient gel state (a polymer network) formed in the early stage of viscoelastic phase separation, resulting in the suppression of polymer diffusion (stress-diffusion coupling). This effect may be relatively easily understood from the analogy to the diffusion in a gel, where the network connectivity suppresses diffusion. The increase of the polymer composition at a certain point $\mathbf{r}$ inevitably stretches the surrounding polymer network due to its connectivity and creates mechanical stress, suppressing the diffusion of polymers toward $\mathbf{r}$. Since the bulk stress is isotropic, it can be expressed by a scalar variable, i.e. $\tilde{\sigma}_\mathrm{B} = \frac{1}{d} \mathrm{Tr} \overleftrightarrow{\boldsymbol{\sigma}}_B$ ($\overleftrightarrow{\boldsymbol{\sigma}}_B = \tilde{\sigma}_\mathrm{B} \overleftrightarrow{\mathbf{I}}$). We assume that the bulk stress also obeys the following Maxwell-type equation:

$$\frac{D}{Dt} \tilde{\sigma}_\mathrm{B} = -\frac{1}{\tau_\mathrm{B}(\phi)} \tilde{\sigma}_\mathrm{B} + G_\mathrm{B}(\phi) \nabla \cdot \mathbf{v}_\mathrm{p}, \tag{11}$$

where τ_B and G_B are the relaxation time and the modulus of the bulk stress, respectively.

Here we emphasise that the crucial viscoelastic effects originate from the steep ϕ-dependence of $G_\mathrm{B}(\phi)$, $\tau_\mathrm{B}(\phi)$, $G_\mathrm{S}(\phi)$ and $\tau_\mathrm{S}(\phi)$. The steep ϕ-dependence of these rheological parameters can be intuitively understood for a polymer solution. Phase separation creates the spatial heterogeneity of $\phi(\mathbf{r}, t)$. In a high ϕ (dense) polymer-rich region, polymer entanglement significantly enhances viscoelasticity, whereas there are few viscoelastic effects for a low ϕ (dilute) solvent-rich region. This steep ϕ-dependence of the viscoelastic properties and the resulting asymmetric stress division is the origin of the 'dynamic asymmetry' between the two phase-separating

phases. A similar strong dynamic asymmetry also exists in any mixtures with the size asymmetry between the components, such as protein solutions, colloid suspensions, emulsions, and membrane systems. We emphasise that if both phases are equally slow and viscoelastic yet without strong dynamic asymmetry, ordinary phase separation should occur, not VPS[36]. The dynamic asymmetry between the two phases is a prerequisite for VPS.

By solving the above set of equations, we can simulate and reproduce pattern formation during viscoelastic phase separation[87,88]. We emphasise that the above set of equations describing VPS provides the most general model of phase separation, which can describe any phase-separation type, including solid, liquid, and viscoelastic ones[35]. For example, if we set $\sigma = 0$, the VPS model reduces to the ordinary phase separation model of liquid mixtures.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Code availability

There are no custom codes used to generate results in the paper.

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Author contributions

H.T. conceived the project and wrote the manuscript.

Competing interests

The authors declare no competing interests.

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